LAPORAN
WORKSHOP DEVELOPER OPERATIONAL
MINGGU KE- 9

MATERI PEMBELAJARAN

Docker Swarm

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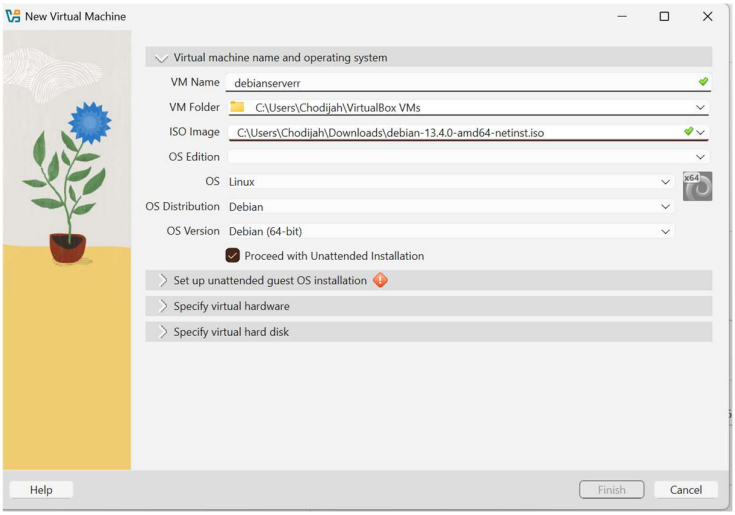
HASIL DAN PEMBAHASAN

1. Langkah pertama install virtual box dan unduh debian di browser
2. Kemudian masuk di virtual box dan buat new VM, pada bagian ISO Image pilih debian yang sudah di unduh.



3. Buat 2 debian untuk dijadikan manager dan worker
4. Masuk sebagai root pada masing-masing debian dan masukkan password
5. Lalu jalankan perintah seperti dibawah ”apt update” yang digunakan untuk memperbarui daftar paket (repository) pada sistem, tunggu hingga selesai

```
vboxuser@debianserver:~$ su -
Password:
root@debianserver:~# apt update
Ign:1 http://security.debian.org/debian-security trixie-security InRelease
Hit:1 http://security.debian.org/debian-security trixie-security InRelease
Hit:2 http://deb.debian.org/debian trixie InRelease
Hit:3 http://deb.debian.org/debian trixie-updates InRelease
All packages are up to date.
root@debianserver:~# apt install docker.io -y
Installing:
  docker.io

Installing dependencies:
  containerd     iptables         libip6tc2                  needrestart
  criu           libcompel1       libmodule-find-perl        patch
  docker-buildx  liberror-perl    libnet1                    python3-protobuf
  docker-cli     libintl-perl     libproc-processtable-perl  python3-pycriu
  git            libintl-xs-perl  libsort-naturally-perl     runc
  git-man        libip4tc2        libterm-readkey-perl       tini

Suggested packages:
  containernetworking-plugins  xfsprogs          git-cvs
  docker-doc                   zfs-fuse          git-mediawiki
  aufs-tools                   | zfsutils-linux  git-svn
```

6. Kemudian setelah selesai, install docker di kedua tempat (debian) dengan perintah “apt install -y docker.io”
7. Lalu start docker dengan perintah “systemctl start docker”
8. Cek ip setiap debian dengan perintah seperti dibawah “ip a”

```
root@debianserver:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1
    link/ether 08:00:27:95:e5:70 brd ff:ff:ff:ff:ff:ff
    altname enx08002795e570
    inet 192.168.1.69/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
       valid_lft 2703sec preferred_lft 2703sec
    inet6 fe80::a00:27ff:fe95:e570/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:53:54:23:a8 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
```

9. Selanjutnya, gunakan ip pada debian pertama untuk melakukan ping ke debian kedua, lalu lakukan juga sebaliknya dari debian kedua ke debian pertama.

```
root@debianserver:~# ping 192.168.1.30
PING 192.168.1.30 (192.168.1.30) 56(84) bytes of data.
64 bytes from 192.168.1.30: icmp_seq=1 ttl=64 time=67.0 ms
64 bytes from 192.168.1.30: icmp_seq=2 ttl=64 time=9.68 ms
64 bytes from 192.168.1.30: icmp_seq=3 ttl=64 time=42.7 ms
64 bytes from 192.168.1.30: icmp_seq=4 ttl=64 time=142 ms
64 bytes from 192.168.1.30: icmp_seq=5 ttl=64 time=50.7 ms
64 bytes from 192.168.1.30: icmp_seq=6 ttl=64 time=55.0 ms
^C
--- 192.168.1.30 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 9.678/61.252/142.454/40.377 ms
```

```
root@deb00:~# ping 192.168.1.69
PING 192.168.1.69 (192.168.1.69) 56(84) bytes of data.
From 192.168.1.30 icmp_seq=1 Destination Host Unreachable
ping: sendmsg: No route to host
From 192.168.1.30 icmp_seq=2 Destination Host Unreachable
From 192.168.1.30 icmp_seq=3 Destination Host Unreachable
From 192.168.1.30 icmp_seq=5 Destination Host Unreachable
From 192.168.1.30 icmp_seq=6 Destination Host Unreachable
From 192.168.1.30 icmp_seq=7 Destination Host Unreachable
64 bytes from 192.168.1.69: icmp_seq=8 ttl=64 time=2363 ms
64 bytes from 192.168.1.69: icmp_seq=9 ttl=64 time=1339 ms
64 bytes from 192.168.1.69: icmp_seq=10 ttl=64 time=314 ms
64 bytes from 192.168.1.69: icmp_seq=11 ttl=64 time=110 ms
64 bytes from 192.168.1.69: icmp_seq=12 ttl=64 time=19.2 ms
64 bytes from 192.168.1.69: icmp_seq=13 ttl=64 time=60.2 ms
64 bytes from 192.168.1.69: icmp_seq=14 ttl=64 time=75.6 ms
64 bytes from 192.168.1.69: icmp_seq=15 ttl=64 time=103 ms
64 bytes from 192.168.1.69: icmp_seq=16 ttl=64 time=12.4 ms
64 bytes from 192.168.1.69: icmp_seq=17 ttl=64 time=91.1 ms
64 bytes from 192.168.1.69: icmp_seq=18 ttl=64 time=12.4 ms
64 bytes from 192.168.1.69: icmp_seq=19 ttl=64 time=81.1 ms
```

10. Jika keduanya sudah berhasil saling terhubung dalam satu jaringan, jalankan perintah berikut untuk memperoleh token dari Debian yang berperan sebagai manager.

```
root@debianserver:~# docker swarm init --advertise-addr 192.168.1.69
Swarm initialized: current node (8i2r69u94ptkcbxygyxm8dkip) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-4qlg1zlai1gnwm4z5cesf97f5qacpx2pqxk2ywx2ptrsl09xug-bfy6vs57
6791x89j2646 192.168.1.69:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

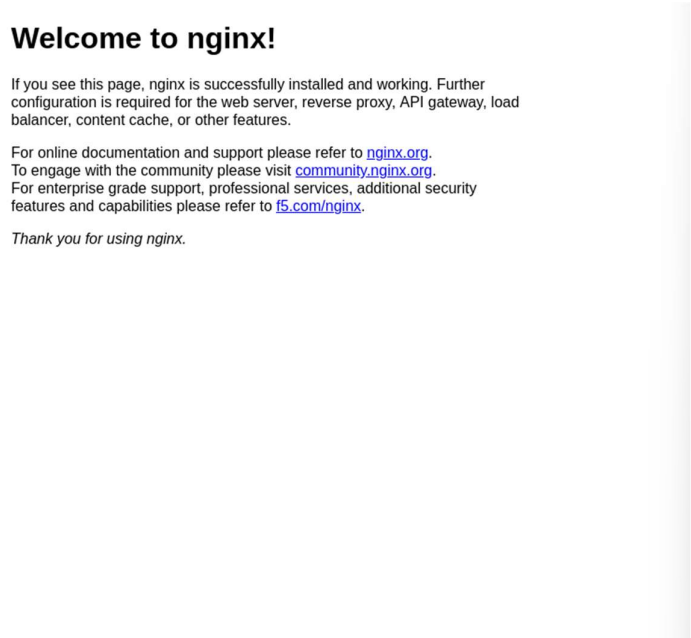
11. Setelah itu, akan muncul perintah docker swarm join .... salin perintah tersebut lalu jalankan pada Docker di node worker. Jika berhasil, akan muncul keterangan “This node joined a swarm as a worker.”

```
root@deb00:~# docker swarm join --token SWMTKN-1-4qlg1zlai1gnwm4z5cesf97f5qacpx2pqx
wx2ptrsl09xug-bfy6vs5742yv56791x89j2646 192.168.1.69:2377
This node joined a swarm as a worker.
```

12. Untuk memastikan proses berhasil, kembali ke debian manager lalu jalankan perintah docker node ls. Jika berhasil, akan terlihat daftar node yang sudah tergabung dalam swarm, termasuk manager dan worker.

```
root@debianserver:~# docker node ls
ID                            HOSTNAME       STATUS    AVAILABILITY   MANAGER STATUS   ENGINE VERS
byoo7yh4jqbami8tbp3kwziry     deb00          Ready     Active                          26.1.5+dfsg
8i2r69u94ptkcbxygyxm8dkip *   debianserver   Ready     Active         Leader           26.1.5+dfsg
```

13. Jalankan layanan nginx pada docker swarm dengan membuat service bernama web dan port 8080 ke 80 menggunakan perintah “docker service create --name web -p 8080:80 nginx”.

```
root@debianserver:~# docker run -d -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
5435b2dcdf5c: Pull complete
054715a6bffa: Pull complete
88d1d984b765: Pull complete
4a038fd18db1: Pull complete
84e114c2bb36: Pull complete
7b5d674621c2: Pull complete
448ea5cac5d5: Pull complete
Digest: sha256:7f0adca1fc6c29c8dc49a2e90037a10ba20dc266baaed0988e9fb4d0d8b85ba0
Status: Downloaded newer image for nginx:latest
714b34f775aededc5fe3416e9971c3a950aace97da75385549ba591828e67a5a
root@debianserver:~#
```

14. Lalu akses di browser dengan perintah url melalui alamat http://IP-Manager:8080